Business Dynamism — Problem Set 1

Firm Dynamics and Growth

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Throughout, the Davis–Haltiwanger–Schuh (DHS) growth rate of unit i between $t-1$ and t is

$$g_{it} \;=\; \frac{E_{it} - E_{it-1}}{\frac{1}{2}\big(E_{it} + E_{it-1}\big)} \;=\; \frac{\Delta E_{it}}{\bar{E}_{it}}, \qquad \bar{E}_{it} \equiv \tfrac{1}{2}\big(E_{it} + E_{it-1}\big), \tag{1}$$

with $E_{it}, E_{it-1} \geq 0$ and $\bar{E}_{it} > 0$ (the unit is active in at least one of the two years). It is convenient to write g in terms of the gross growth factor $r \equiv E_{it}/E_{it-1}$,

$$g_{it} \;=\; 2\,\frac{E_{it} - E_{it-1}}{E_{it} + E_{it-1}} \;=\; 2\,\frac{r-1}{r+1}. \tag{2}$$

1 DHS growth: bounds, symmetry, log approximation

(a) Bounded in $[-2, +2]$; entry and exit are the endpoints

Let $s = E_{it} + E_{it-1} > 0$ and $d = E_{it} - E_{it-1}$. Because both employment levels are non-negative, $|d| \leq s$, so from (1)

$$|g_{it}| = 2\,\frac{|d|}{s} \leq 2.$$

The bound is attained exactly at the two margins:

$$g_{it} = +2 \iff E_{it-1} = 0 \quad (\textit{entrant}\text{: born in } t), \qquad g_{it} = -2 \iff E_{it} = 0 \quad (\textit{exit}\text{: dies in } t).$$

Equivalently, in (2) the map $r \mapsto 2(r-1)/(r+1)$ is strictly increasing on $r \in [0, \infty)$, with $g \to -2$ as $r \to 0$ (exit) and $g \to +2$ as $r \to \infty$ (entry). The economic content is that the average-size denominator $\bar{E}_{it}$ never vanishes for an active unit, so the *same* formula (1) assigns a finite rate to entrants ($+2$) and exiters (-2). The conventional base-year rate $\Delta E_{it}/E_{it-1}$ is undefined for entrants and equals -1 for exiters, and the log difference diverges at both margins; the DHS rate avoids both problems, which is precisely why entry and exit "fit inside" one measure.

(b) Symmetry of expansions and contractions

A proportional expansion by factor r and the reciprocal contraction by $1/r$ produce growth rates of equal magnitude and opposite sign. From (2),

$$g(1/r) = 2\,\frac{1/r - 1}{1/r + 1} = 2\,\frac{1-r}{1+r} = -\,g(r).$$

Thus g is antisymmetric under $E_{it} \leftrightarrow E_{it-1}$ (time reversal). The two leading cases:

$$\text{doubling } (r = 2):\quad g = 2 \cdot \tfrac{1}{3} = \tfrac{2}{3}, \qquad \text{halving } (r = \tfrac{1}{2}):\quad g = 2 \cdot \tfrac{-1/2}{3/2} = -\tfrac{2}{3}.$$

For contrast, the base-year rate gives $+1$ for a doubling and $-\frac{1}{2}$ for a halving: it treats gains and equal-proportion losses asymmetrically, whereas the DHS rate does not.

(c) Second-order agreement with the log first difference

Write the log change as $\ell \equiv \Delta \ln E_{it} = \ln r$, so $r = e^{\ell}$. Substituting into (2) and multiplying through by $e^{-\ell/2}$,

$$g_{it} = 2\,\frac{e^{\ell}-1}{e^{\ell}+1} = 2\,\frac{e^{\ell/2}-e^{-\ell/2}}{e^{\ell/2}+e^{-\ell/2}} = 2\tanh\!\Big(\tfrac{\ell}{2}\Big). \tag{3}$$

Expanding $\tanh x = x - \frac{x^3}{3} + O(x^5)$ at $x = \ell/2$,

$$g_{it} = \ell - \frac{\ell^3}{12} + O(\ell^5) = \Delta \ln E_{it} - \frac{1}{12}\big(\Delta \ln E_{it}\big)^3 + O(\ell^5). \tag{4}$$

Both g and ℓ are odd functions of ℓ, so their quadratic terms vanish identically and they coincide through second order; the first discrepancy is cubic, $g - \Delta \ln E = -\frac{1}{12}(\Delta \ln E)^3 + O(\ell^5)$. Hence around zero the DHS rate is a second-order approximation to the log first difference, $g_{it} = \Delta \ln E_{it} + O(\ell^3)$. (Equivalently $\Delta \ln E_{it} = 2\,\mathrm{artanh}(g/2) = g + g^3/12 + \cdots$.) The vanishing of the even-order terms is exactly the antisymmetry established in (b), so parts (b) and (c) are two faces of the same property.

2 The share-weighted aggregate is the aggregate DHS rate

Let $i = 1, \ldots, N$ index the firms in a given economy (or sector) in year t, each with rate $g_i = \Delta E_i/\bar{E}_i$. Define the DHS *activity weight* as the firm's share of aggregate average employment,

$$z_i \;=\; \frac{\bar{E}_i}{\sum_j \bar{E}_j}, \qquad \bar{E} \equiv \sum_j \bar{E}_j = \tfrac{1}{2}\sum_j \big(E_{jt} + E_{jt-1}\big), \qquad z_i \geq 0,\;\; \sum_i z_i = 1. \tag{5}$$

The share-weighted sum of firm growth rates is $G \equiv \sum_i z_i g_i$.

(a) Equals net jobs over average employment

Because $g_i = \Delta E_i/\bar{E}_i$, the firm-specific denominator cancels the weight:

$$G = \sum_i z_i\, g_i = \sum_i \frac{\bar{E}_i}{\sum_j \bar{E}_j}\cdot\frac{\Delta E_i}{\bar{E}_i} = \frac{\sum_i \Delta E_i}{\sum_j \bar{E}_j} = \frac{\overbrace{\sum_i \big(E_{it} - E_{it-1}\big)}^{\text{net jobs created/destroyed}}}{\underbrace{\tfrac{1}{2}\sum_i \big(E_{it} + E_{it-1}\big)}_{\text{average employment}}}. \tag{6}$$

The numerator is economy-wide net job creation between $t-1$ and t (in levels); the denominator is economy-wide average employment over the two years. This is the required identity, and it is the cancellation of $\bar{E}_i$ that makes the average-size weight (5)—rather than a base-year weight—the correct one.

(b) Equals the employment-weighted mean of firm rates

By construction $G = \sum_i z_i g_i$ with weights $z_i \geq 0$ summing to one: it is therefore the (average-)employment-weighted mean of the firm growth rates. Two consequences follow. First, G inherits the bound of part 1(a): as a convex combination of rates in $[-2, 2]$, $G \in [-2, 2]$. Second, the aggregation is *internally consistent.* Writing $E_t^{\text{agg}} = \sum_i E_{it}$, the numerator of (6) is $E_t^{\text{agg}} - E_{t-1}^{\text{agg}}$ and the denominator is $\frac{1}{2}(E_t^{\text{agg}} + E_{t-1}^{\text{agg}})$, so

$$G = \frac{E_t^{\text{agg}} - E_{t-1}^{\text{agg}}}{\tfrac{1}{2}\big(E_t^{\text{agg}} + E_{t-1}^{\text{agg}}\big)},$$

i.e. the employment-weighted mean of firm DHS rates is exactly the DHS rate computed on aggregate employment. Entrants ($\bar{E}_i = \frac{1}{2}E_{it}$, $g_i = 2$) and exiters ($\bar{E}_i = \frac{1}{2}E_{it-1}$, $g_i = -2$) enter this mean with weight proportional to their own average size and contribute $z_i g_i = \pm E_i/\bar{E}$ jobs per unit of average employment, so the identity absorbs the margins of part 1(a) without special treatment.

Numerical check. Three firms over one year: a continuer ($E_{t-1} = 100, E_t = 120$), an exiter ($100 \to 0$), and an entrant ($0 \to 40$). Then $\bar{E} = (110 + 50 + 20) = 180$, and

$$g_1 = \tfrac{20}{110},\ g_2 = -2,\ g_3 = 2; \qquad \sum_i z_i g_i = \frac{110}{180}\tfrac{20}{110} + \frac{50}{180}(-2) + \frac{20}{180}(2) = \frac{20 - 100 + 40}{180} = -\frac{40}{180} = -0.2\bar{2},$$

which equals $\sum_i \Delta E_i/\bar{E} = (20 - 100 + 40)/180$, confirming (6).

3 Reproducing Figure 1 of Haltiwanger et al. (2013)

Target. Figure 1 of Haltiwanger et al. (2013) reports, for 1992–2005, the share of employment, of gross job creation (JC) and of gross job destruction (JD) accounted for by six firm groups: the cross of firm size—*small* (< 500) versus *large* (≥ 500)—with firm type—*startups* (firm age 0), *young* (ages 1–9), and *mature* (ages 10+). Job flows are measured at the establishment level but classified by the size and age of the *parent firm*.

Data. U.S. Census Bureau Business Dynamics Statistics (BDS), national *Firm Age by Firm Size* table (variables `emp`, `job_creation`, `job_destruction` by `fage` × `fsize` × year).[1] Firm age 0 is a firm all of whose establishments are entrants, so its JC is firm births and its JD is structurally zero (Census Bureau, 2025b).

Construction. Firm size is aggregated to small/large at the 500-employee boundary and firm age to the three types above (the left-censored age class, i.e. firms born before 1977, is treated as mature). For each of the six groups the annual levels of employment, JC and JD are pooled over 1992–2005 and expressed as a percentage of the corresponding economy-wide total, giving the eighteen shares plotted. The firm-level growth aggregation underlying these flows is the identity of Section 2: within a size–age cell, the cell's contribution to the aggregate net rate is its share-weighted DHS rate, and JC and JD are the gross components of that flow.

Cell suppression. BDS flags cells with too few firms (disclosure, `D`), unreliable time series (`S`), structurally missing cells (`X`), or non-calculable rates (`N`); flagged cells carry a data value of 0 (Census Bureau, 2025a). These are set to missing rather than read as zeros so that suppressed cells neither inflate a group total nor its denominator. At the national level the size–age cells are large and suppression is negligible; the safeguard matters mainly if the exercise is repeated at finer industry or geographic detail.

Result. Two features of Figure 1 carry the paper's message. First, shares of JC and JD are broadly proportional to shares of employment: large, mature firms hold the most jobs ($\approx 45\%$) and therefore create and destroy the most. Second, young firms and startups are *disproportionately* represented in the flows relative to their employment: startups (about 3% of employment) account for a much larger share of JC and essentially none of JD, and young firms contribute more to both JC and JD than their employment share—the "up-or-out" churning that motivates the paper's focus on age rather than size.

[1] Downloaded from census.gov; the paper-era release file `bds_f_agesz_release.csv` (1977–2014) and the current `bds2023_fa_fz.csv` vintage both reproduce the qualitative pattern.

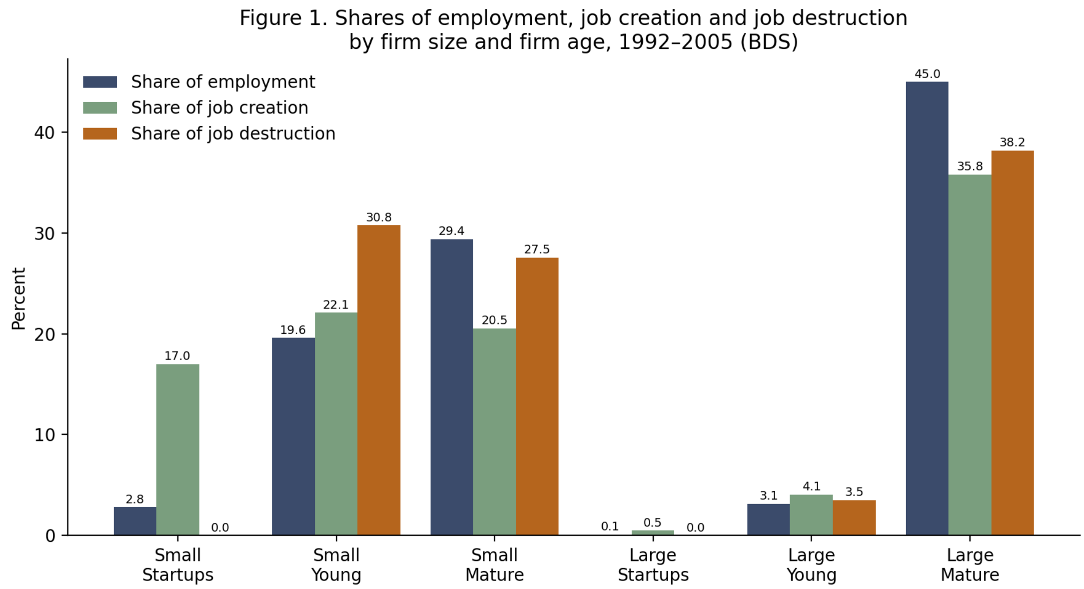

Figure 1: Shares of employment, gross job creation and gross job destruction by firm size and firm age, BDS, 1992–2005. Generated by `reproduce_figure1.py` (Appendix A)

A Reproduction code

The full script (`reproduce_figure1.py`) downloads the BDS table, handles suppression flags, maps cells to the six groups, pools 1992–2005, and writes the figure. Core logic:

```
SUPPRESSION_FLAGS = {"D", "N", "S", "X"}   # flagged cells -> missing, not 0

def size_class(label):                    # Small (<500) vs Large (>=500)
    lo = int(re.findall(r"\d+", label)[0])
    return "Large" if lo >= 500 else "Small"

def age_class(label):                     # Startups / Young (1-9) / Mature
    if "left␣censored" in label.lower(): return "Mature"
    lo = int(re.findall(r"\d+", label)[0])
    return "Startups" if lo == 0 else ("Young" if lo <= 9 else "Mature")

# pool levels over 1992-2005, then form within-year-total shares
lev = d.groupby(["size_g","age_g"])[["emp","jc","jd"]].sum(min_count=1)
for m in ("emp","jc","jd"):
    lev[f"{m}_share"] = 100 * lev[m] / lev[m].sum()
```

Use of AI tools. Claude (Anthropic) was used to check the algebra in Sections 1–2 and to draft and debug the reproduction script in Section 3; all derivations and the final text were verified by the author.

References

U.S. Census Bureau (2025a). *Business Dynamics Statistics: Codebook and Glossary (2023 Release)*. Center for Economic Studies.

U.S. Census Bureau (2025b). *Business Dynamics Statistics: Methodology*. Center for Economic Studies.

Davis, S. J., Haltiwanger, J., and Schuh, S. (1996). *Job Creation and Destruction.* MIT Press, Cambridge, MA.

Haltiwanger, J., Jarmin, R. S., and Miranda, J. (2013). Who creates jobs? Small versus large versus young. *Review of Economics and Statistics*, 95(2):347–361.